

Self-attention, from first principles

Match, mix, project, add: a token keeps its row and adds what it read from the allowed earlier tokens.

THE QUESTION

How can a token pull in the context it needs, when which earlier words matter changes from sentence to sentence?

THE INTUITION

Each token asks with a query; every allowed source offers a key and a value. Query-key matches choose the weights, the weights mix the values, and the result is projected and added to the token's own row.

KEY TAKEAWAY

Q and K choose where to read; V decides what is sent. The update Δe is added to e , and a causal mask keeps the future out.

e current row $\rightarrow q \cdot k \cdot v \rightarrow$ weights $\alpha \rightarrow$ message $m \rightarrow$ update $\Delta e \rightarrow e + \Delta e$

1 Three roles from one row

e_7 bank

water	finance	person	glue
0.70	0.70	0.10	0.80

same row, three learned maps:
 $q = e \cdot W_Q$ $k = e \cdot W_K$ $v = e \cdot W_V$

water? = $0.70 \times 1 + 0.80 \times 0.7 = 1.26$
 who? = $0.10 \times 0.4 = 0.04$

values rounded; axis names are teaching labels

$\times W_Q (4 \times 3)$ q_7 $\times W_K (4 \times 3)$ k_7 $\times W_V (4 \times 2)$ v_7

water?	finance?	who?
1.26	1.26	0.04

water?	finance?
0.70	0.70

water	finance
0.70	0.70

Query q_i : what the receiving token looks for. **Key** k_j : what source j can be matched on. **Value** v_j : what source j sends if it is read. All three are linear maps of the same row.

Directory analogy: query "contact the library" matches key "Library"; the value {ext. 204, room 2B, 09:00-18:00} is what you get. The name does not supply the details. Real q, k, v are learned vectors, not records.

2 Match, then mix: bank reads its prefix

source	key k	$s = q \cdot k / \sqrt{3}$	weight α	value v	$\alpha \cdot v$
The	[0.10, 0.00, 0.10]	0.08	0.050	[0.10, 0.00]	[0.01, 0.00]
fisherman	[2.00, 0.10, 2.10]	1.58	0.226	[2.00, 0.10]	[0.45, 0.02]
sat	[0.30, 0.00, 0.60]	0.23	0.059	[0.30, 0.00]	[0.02, 0.00]
beside	[0.60, -0.10, 0.10]	0.37	0.067	[0.60, -0.10]	[0.04, -0.01]
the	[0.10, 0.10, 0.00]	0.15	0.054	[0.10, 0.10]	[0.01, 0.01]
river	[3.10, -0.10, 0.00]	2.18	0.414	[3.10, -0.10]	[1.28, -0.04]
bank	[0.70, 0.70, 0.10]	1.02	0.130	[0.70, 0.70]	[0.09, 0.09]

query $q_7 = [1.26, 1.26, 0.04]$ sum = 1 $m_7 = [1.89, 0.07]$

Score each allowed source, normalize over $j \leq i$, then mix:

$$s_{ij} = \frac{q_i k_j^T}{\sqrt{d_k}}, \quad \alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{r \leq i} \exp(s_{ir})}, \quad m_i = \sum_{j \leq i} \alpha_{ij} v_j.$$

Weights are nonnegative and each row sums to 1; they depend on Q and K, not V. One scalar α_{ij} scales every coordinate of v_j . Hard retrieval returns one winner's value; soft attention mixes. River matches best: $\alpha_{7,6} = 0.41$.

5 The same steps in PyTorch

```
q = E[6:7] @ W_Q # [1, 3]
K, V = E[:7] @ W_K, E[:7] @ W_V # tokens 1-7
scores = q @ K.T / math.sqrt(K.shape[-1])
alpha = scores.softmax(dim=-1) # [1, 7]
m = alpha @ V # [1, 2]
e_new = E[6:7] + m @ W_O # [1, 4]
```

3 Project the message, then add it

m_7 (value space, 2) Δe_7 (embedding space, 4)

1.89	0.07
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$2 \rightarrow 4$

W_O

1.51	-0.30	0.58	0.18
------	-------	------	------

e_7 (kept unchanged) $\oplus$

0.70	0.70	0.10	0.80
------	------	------	------

$e'_7 = e_7 + \Delta e_7$

2.21	0.40	0.68	0.98
------	------	------	------

water finance person glue

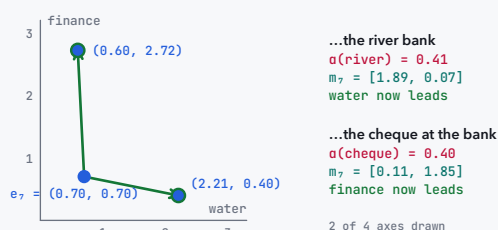
A 2-number message cannot be added to a 4-number row, so a learned linear map changes its width:

$$W_O = \begin{pmatrix} 0.8 & -0.2 & 0.3 & 0.1 \\ -0.1 & 1.1 & 0.2 & -0.2 \end{pmatrix}, \quad \Delta e_7 = m_7 W_O, \\ e'_7 = e_7 + \Delta e_7.$$

It mixes both inputs into all four outputs: not zero-padding, copying, or an orthogonal projection. The residual route keeps e_7 untouched until the addition; water (2.21) now leads.

Δe is the lesson's teaching name for the attention output; $e + \Delta e$ is the standard residual connection. A bias is possible; this toy omits it.

4 Same bank, different context



bank starts from the same row in both sentences (half water, half finance) and reads only tokens 1-7. River pulls it toward water, cheque toward finance: $e'_7 = [2.21, 0.40, 0.68, 0.98]$ vs $[0.60, 2.72, 0.50, 0.44]$.

Hand-chosen toy: $d_{\text{model}} = 4$ (water, finance, person, glue), $d_k = 3$, $d_v = 2$. Named axes are teaching aids, not a claim about trained models. Values rounded.

6 Pause and think

All scores equal? Every weight is $1/7$, so m_7 is the plain mean of the values, $[0.99, 0.10]$.

Is the query the updated bank? No. Within a layer every q, k, v comes from the layer's input rows; e'_7 feeds the next layer.

Why keep e ? The update adds context; the residual keeps what the token already was.

Where Q, K and V come from

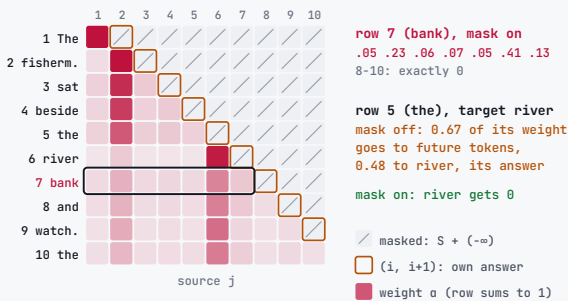
Row-vector convention · T token positions · one attention head · biases omitted in the attention equations.

7 Same matching width, different jobs

Each current row supplies all three: $q_i = e_i W_Q$, $k_i = e_i W_K$, $v_i = e_i W_V$. Q and K must share width d_k for the dot product; equal shape does not require equal entries or equal maps, and d_v need not equal d_k .

For “Maya grabbed a coat because she was cold,” a query from she may ask for a person; Maya’s key may advertise a person-like source; its value may send identity and number. W_Q extracts what a token seeks, W_K what it can match, W_V what it sends. Tying them is possible but restricts the roles. Each matrix is learned, shared across positions, and fixed during a forward pass.

8 A causal mask blocks the answer



At position i we predict token $i + 1$, so i may read $j \leq i$ (itself included), never the future:

$$M_{ij} = \begin{cases} 0 & \text{if } j \leq i \\ -\infty & \text{if } j > i \end{cases}, \quad A = \text{softmax}_{\text{row}}(S + M).$$

Since $\exp(-\infty) = 0$, future values get exactly zero weight. Mask before softmax: zeroing afterwards without renormalizing breaks the rows, and setting scores to 0 blocks nothing ($e^0 = 1$). Example: $[1, 2, 3, 4] + [0, 0, -\infty, -\infty] \rightarrow [0.269, 0.731, 0, 0]$.

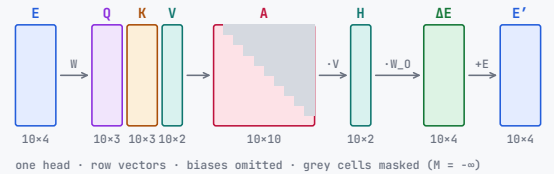
Training: the whole sentence is there, but no position may see its own answer. Generation: the future does not exist yet. The same mask covers both cases.

9 Why not a fixed rule?

	The	fisherm.	sat	beside	the	river	bank	and	watch.	the
mean	.10	.10	.10	.10	.10	.10	.10	.10	.10	.10
decay $0.7^{(t-j)}$	.01	.02	.03	.04	.05	.07	.11	.15	.22	.31
attention, A	.03	.20	.04	.04	.03	.45	.10	.03	.05	.03
attention, B	.03	.24	.03	.42	.03	.03	.10	.03	.05	.03
				cheque (B)		river (A)				

A mean or a positional decay picks weights before reading the words, so both sentences get the same row. Only attention recomputes its weights from content: content-dependent soft retrieval.

10 The complete matrix path

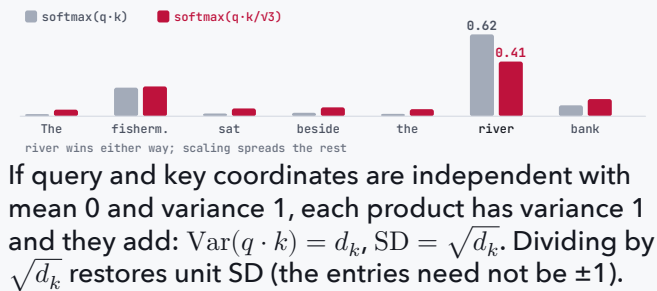


$$S = \frac{QK^T}{\sqrt{d_k}}, \quad A = \text{softmax}_{\text{row}}(S + M),$$

$$\Delta E = (AV)W_O, \quad E' = E + \Delta E.$$

Object	Shape	Object	Shape
$E, \Delta E, E'$	$T \times d_{\text{model}}$	W_Q, W_K	$d_{\text{model}} \times d_k$
Q, K	$T \times d_k$	W_V	$d_{\text{model}} \times d_v$
V, AV	$T \times d_v$	W_O	$d_v \times d_{\text{model}}$
S, M, A	$T \times T$	toy	10; 4, 3, 2

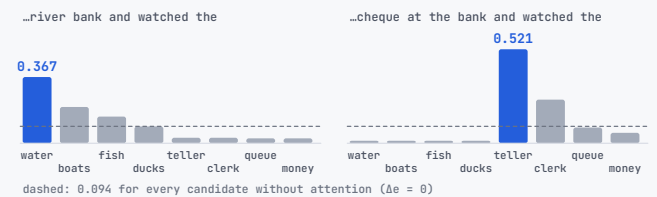
11 Scaling by $\sqrt{d_k}$ controls score spread



If query and key coordinates are independent with mean 0 and variance 1, each product has variance 1 and they add: $\text{Var}(q \cdot k) = d_k$, $\text{SD} = \sqrt{d_k}$. Dividing by $\sqrt{d_k}$ restores unit SD (the entries need not be ± 1).

3,000 simulated pairs per row. With $d_k = 64$, scores $[-8, 8]$ give $p_A \approx 0.000000113$; divided by 8, $[-1, 1]$ gives $p_A \approx 0.1192$. Scaling avoids needlessly sharp weights and tiny softmax gradients. Overflow is separate: subtracting the max is stable but keeps the gaps.

12 From Δe to the next token



The final the asks the same broad query in both sentences, $q_{10} = [1.61, 1.61, 0.00]$; the keys and values supply the scene. Its row e'_{10} feeds a small head: $h = \text{ReLU}(e'_{10}W_1 + b_1)$, $z = hW_2 + b_2$, $p = \text{softmax}(z)$ with widths $4 \rightarrow 8 \rightarrow 20$. Water 0.367 vs teller 0.521: both plausible, neither the only answer. This toy head is not the Transformer’s feed-forward block.

Q and K choose the weights · the weights mix $V \cdot W_O$ maps the message back · add Δe to e · mask the future

Numbers from the hand-designed toy in the Part 2 interactive lesson (not a trained language model; rounded for display). Vaswani et al., *Attention Is All You Need* (2017), combines attention with feed-forward layers, position information, normalization and residual connections. Watch the summary video of this part.